

Ashmita Dua

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Education

University of California, Santa Cruz | Undergraduate | Sept, 2024 - June, 2028

Bachelor of Science (B.Sc.) in Computer Science | GPA 3.66/4.0 (Dean's Honor List Winter 2025)

Skills / Tech Stacks

- **Game Development:** Unity/C#, Procedural Content Generation (Wave Function Collapse), Level Design
- **AI/ML:** LLMs (GPT-4, Gemini, Claude), Prompt Engineering, RAG Systems, PyTorch, CNNs, NLP
- **Programming Languages:** Python, C/C++, JavaScript, HTML/CSS
- **Tools & Frameworks:** Git, LangChain, OpenCV, APIs, Streamlit

Experience

Augmented Design Lab

September, 2025 - Present

Undergraduate Researcher

- **Reduced LLM validation loop errors by 42%** (78.7% → 46.3%) in the level generation system by redesigning system prompts to enforce action-first behavior and structured tool use, in Pewter Platformer (See Projects section)
- Led a study focused on generating game levels through LLMs/PCG and hierarchical wave function collapse
- Implemented action-first behavioral constraints, reducing repetitive agent loops across **30+ test scenarios**

Sowiz

June, 2025 - July, 2025

AI/ML Intern

- Engineered RAG workflows using LangChain to create an intelligent agent system for dealer support and product knowledge
- Designed and implemented gamified sales platform with a milestone-based reward system to incentivize unit sales performance
- Integrated AI-powered information retrieval system enabling dealers to access real-time product specs and pricing
- Improved dealer conversion rates by providing comprehensive, accessible resources through an automated workflow system

Artificial Intelligence Explainability Accountability Lab

April, 2025 - June, 2025

Undergraduate Researcher

- Led a study focused on LLM safety, logic translation, and inconsistency detection
- Configured local LLM API environments and integrated SWI-Prolog for logical reasoning verification
- Built a knowledge base and curated UCSC-specific questions

Projects

Woven Mood-Based Storytelling LLM | Demo: duashmita.github.io/woven/ | Repo: bit.ly/woven-intoyou

- Designed LLM-powered narrative system guiding users from current emotion to target mood across 8-10 turn interactive stories
- **Achieved 87% emotional arc accuracy** by implementing rule-based validation with a 10-emotion NLP classifier
- Compared GPT-4 vs Gemini performance: Gemini averaged within 2 mood-indices of target; GPT-4 remained 1-3 indices off

HabPet | Demo: bit.ly/HabpetYTDemo | Web: habit-bot.vercel.app | Repo: github.com/Sanyab8/habit-bot

- Built an IoT habit tracker combining Arduino hardware with a web app featuring AI-powered habit detection via OpenCV and progressive milestone rewards
- Engineered Node.js bridge for real-time Arduino-web communication using sensors and actuators (servos, LEDs, buzzers)
- Deployed full-stack application using React/Vite/Zustand on the Vercel platform

Pewter Platformer | Demo: bit.ly/3NVYIYt

- Diagnosed an LLM failure mode where the agent entered repetitive validation loops during level creation
- Iterated the system prompt + interaction flow to enforce 'action-first' behavior and structured tool use
- Benchmarked over 30 test prompts: loop-rate **78.7% → 46.3%** in a **HTML/JavaScript**-based level generator

CopCuts, Barber LLM-based NPCs Unity Game

- Built AI detective game with LLM-powered NPCs featuring dynamic conversations, memory systems, and personality adaptation
- Implemented suspicion mechanics and relationship tracking where NPC responses vary based on player interrogation strategy

Recognition and Certificate

Goldman Sachs Possibilities Summit 2025 | Selected Delegate | Jan 2025

1 of 100 students selected nationally for leadership and career development program